Numerical Methods
Week 13
Ordinary Differential Equations (ODEs)

ODE Outline

- Introduction to ODEs
- Taylor series methods
- Midpoint and Heun’s method
- Runge-Kutta methods

Midpoint Method

Learning Objectives of Lesson 3

- To be able to solve first order differential equations using the Midpoint Method.
- To be able to solve first order differential equations using the Heun's Predictor Corrector Method.

Topic 8: Lesson 3

Lesson 3: Midpoint Methods

- Review Euler Method
- Midpoint Method

Euler Method

Problem

$$\dot{y}(x) = f(x, y)$$

$$y(x_0) = y_0$$

Euler Method

$$y_0 = y(x_0)$$

$$y_{i+1} = y_i + h\ f(x_i, y_i)$$

$$for\ i = 1,2,\ldots$$

Local Truncation Error $O(h^2)$

Global Truncation Error $O(h)$

Introduction

Problem to be solved is a first order ODE :

$$\dot{y}(x) = f(x, y), \qquad y(x_0) = y_0$$

- The methods proposed in this lesson have the general form:

  $$y_{i+1} = y_i + h\,\phi$$

- For the case of Euler: $\phi = f(x_i, y_i)$
- Different forms of ϕ will be used for the Midpoint and Heun's Methods.

Midpoint Method

Problem

$$\dot{y}(x) = f(x, y)$$

$$y(x_0) = y_0$$

Midpoint Method

$$y_0 = y(x_0)$$

$$y_{i+\frac{1}{2}} = y_i + \frac{h}{2} f(x_i, y_i)$$

$$y_{i+1} = y_i + h\ f(x_{i+\frac{1}{2}}, y_{i+\frac{1}{2}})$$

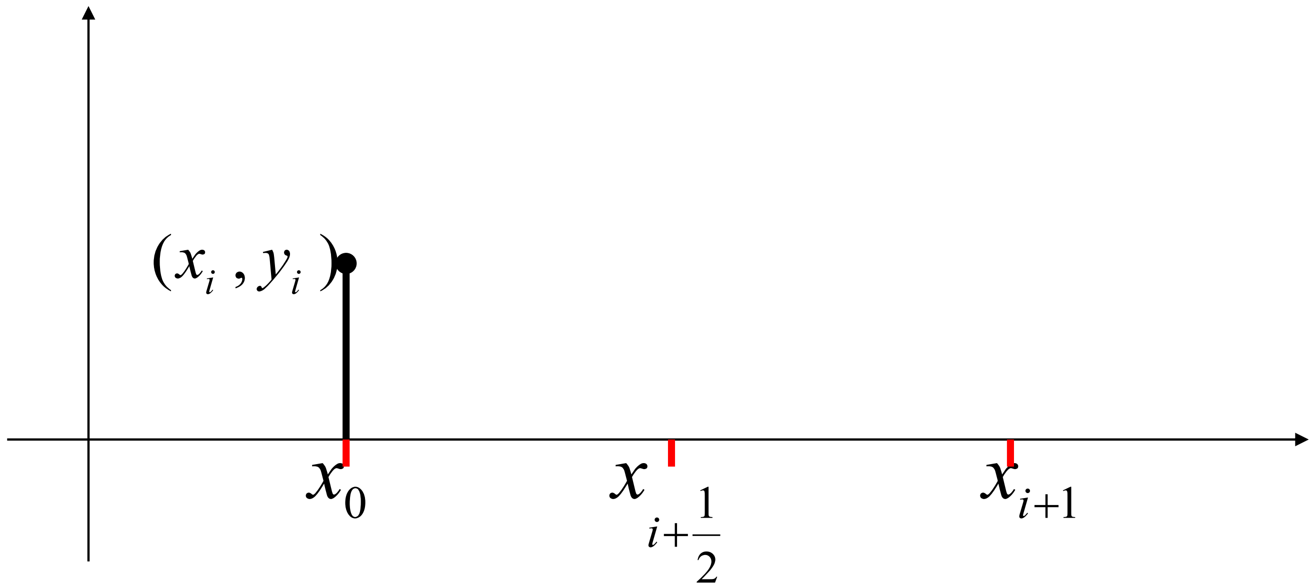
Local Truncation Error $O(h^3)$

Global Truncation Error $O(h^2)$

Motivation

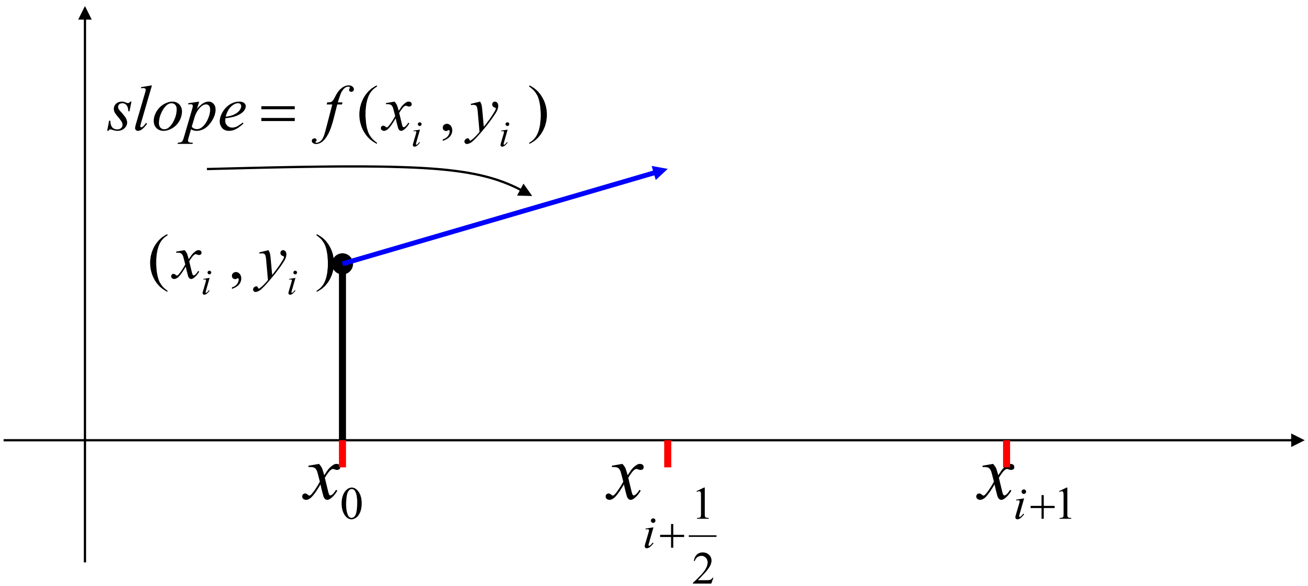
- The midpoint can be summarized as:
  - Euler method is used to estimate the solution at the midpoint.
  - The value of the rate function $f(x,y)$ at the mid point is calculated.
  - This value is used to estimate y_{i+1}.
- Local Truncation error of order $O(h^3)$.
- Comparable to Second order Taylor series method.

Midpoint Method



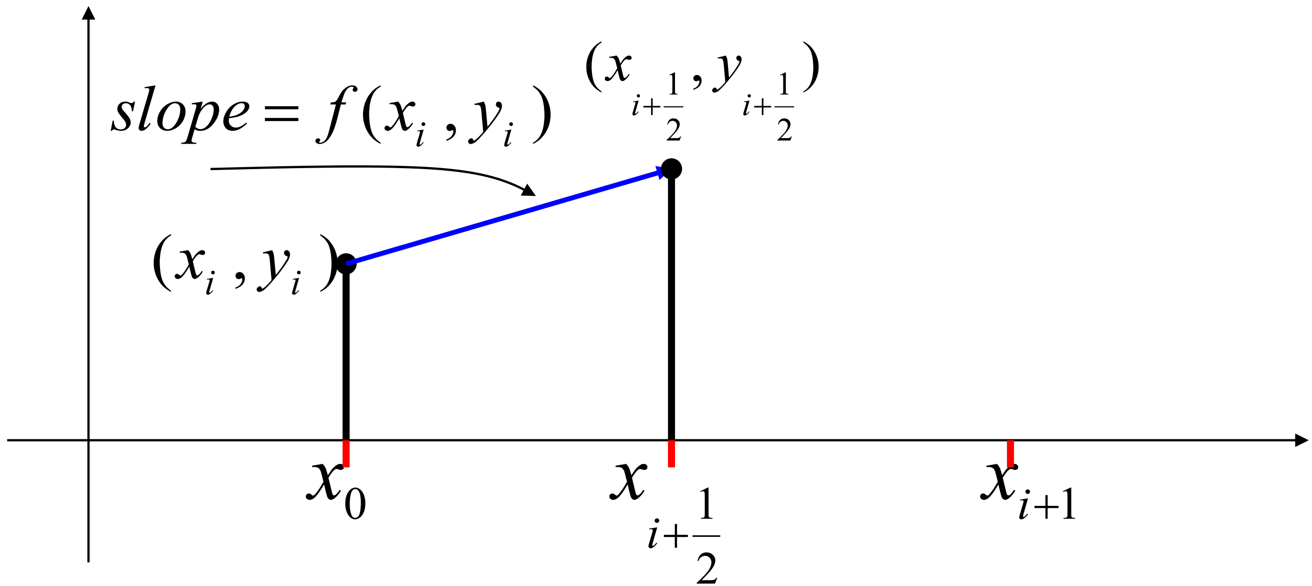
$$y_{i+\frac{1}{2}} = y_i + \frac{h}{2} f(x_i, y_i), \qquad y_{i+1} = y_i + h\, f(x_{i+\frac{1}{2}}, y_{i+\frac{1}{2}})$$

Midpoint Method



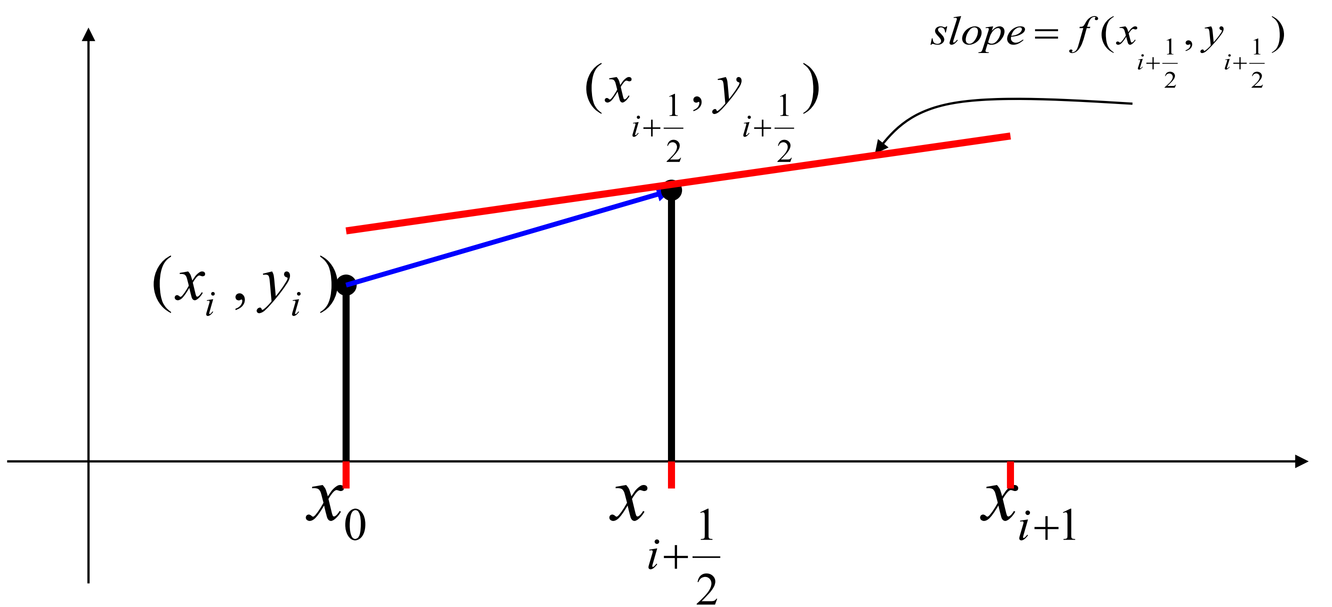
$$y_{i+\frac{1}{2}} = y_i + \frac{h}{2} f(x_i, y_i), \qquad y_{i+1} = y_i + h\, f(x_{i+\frac{1}{2}}, y_{i+\frac{1}{2}})$$

Midpoint Method



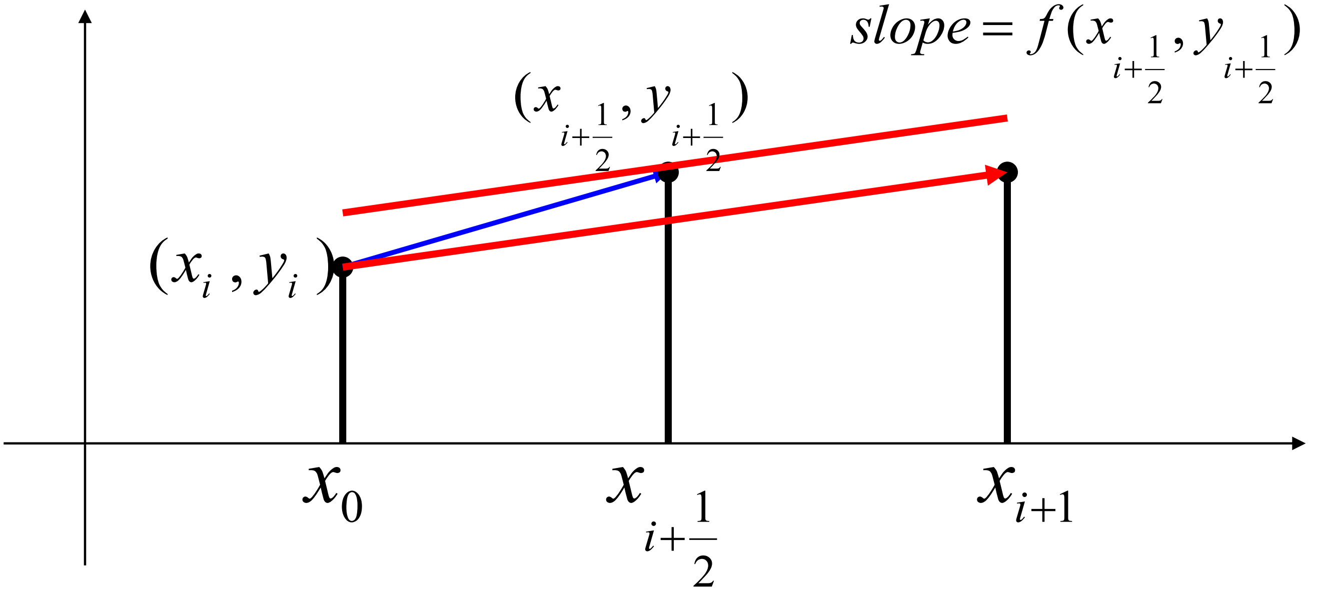
$$y_{i+\frac{1}{2}} = y_i + \frac{h}{2} f(x_i, y_i), \qquad y_{i+1} = y_i + h\, f(x_{i+\frac{1}{2}}, y_{i+\frac{1}{2}})$$

Midpoint Method



$$y_{i+\frac{1}{2}} = y_i + \frac{h}{2} f(x_i, y_i), \qquad y_{i+1} = y_i + h\, f(x_{i+\frac{1}{2}}, y_{i+\frac{1}{2}})$$

Midpoint Method



$$y_{i+\frac{1}{2}} = y_i + \frac{h}{2} f(x_i, y_i), \qquad y_{i+1} = y_i + h\, f(x_{i+\frac{1}{2}}, y_{i+\frac{1}{2}})$$

Example 1

Use the Midpoint Method to solve the ODE

$$\dot{y}(x) = 1 + x^2 + y$$

$$y(0) = 1$$

Use $h = 0.1$. Determine y(0.1) and y(0.2)

Example 1

$$\text{Problem}: f(x,y) = 1 + y + x^2,\ y_0 = y(x_0) = 1, h = 0.1$$

$$\text{Step 1}:$$

$$y_{0+\frac{1}{2}} = y_0 + \frac{h}{2} f(x_0, y_0) = 1 + 0.05(1 + 0 + 1) = 1.1$$

$$y_1 = y_0 + h\ f(x_{0+\frac{1}{2}}, y_{0+\frac{1}{2}}) = 1 + 0.1(1 + 0.0025 + 1.1) = 1.2103$$

$$\text{Step 2}:$$

$$y_{1+\frac{1}{2}} = y_1 + \frac{h}{2} f(x_1, y_1) = 1.2103 + .05(1 + 0.01 + 1.2103) = 1.3213$$

$$y_2 = y_1 + h\ f(x_{1+\frac{1}{2}}, y_{1+\frac{1}{2}}) = 1.2103 + 0.1(2.3438) = 1.4446$$

Summary

- Euler and Midpoint methods are similar in the following sense:

$$y_{i+1} = y_i + h \times slope$$

  - Different methods use different estimates of the slope.
- Midpoint method is comparable in accuracy to the second order Taylor series method.

Comparison

Method		Local truncation error	Global truncation error
Euler Method	$y_{i+1} = y_i + h\ f(x_i, y_i)$	$O(h^2)$	$O(h)$
Heun's Method	Predictor: $y_{i+1}^{0} = y_i + h\ f(x_i, y_i)$ Corrector: $y_{i+1}^{k+1} = y_i + \frac{h}{2}\left(f(x_i, y_i) + f(x_{i+1}, y_{i+1}^{k})\right)$	$O(h^3)$	$O(h^2)$
Midpoint	$y_{i+\frac{1}{2}} = y_i + \frac{h}{2}\ f(x_i, y_i)$ $y_{i+1} = y_i + h\ f(x_{i+\frac{1}{2}}, y_{i+\frac{1}{2}})$	$O(h^3)$	$O(h^2)$